BUSINESS PERFORMANCE REVIEW

Superstore Sales & Profitability Report

A four-year analysis of sales, profit, regions, categories and customer segments (2011–2014)

Prepared using SQL, Python and Power BI

Dataset: Sample Superstore | 9,994 order lines | 5,009 orders | 793 customers

Period covered: Jan 2011 – Dec 2014

Executive Summary

Between January 2011 and December 2014, the Superstore generated **$2.30M** in total sales and **$286K** in profit, an overall profit margin of **12.5%**. Growth was strong and consistent after an early dip: sales fell 2.8% in 2012 before accelerating to +29.3% in 2013 and +20.6% in 2014, closing the period at $734K in annual sales. Profitability, however, has not kept pace with sales growth — margin has drifted down slightly since its 2013 peak, driven largely by heavy discounting and structural losses in the **Furniture** category.

$2.30M	$286K	5,009	$458.6
TOTAL SALES	TOTAL PROFIT	TOTAL ORDERS	AVG ORDER VALUE
4-yr CAGR +14.8%	**12.5% margin**	**793 unique customers**	**1,841 unique products**

Key Takeaways

- **Technology** is the profit engine of the business: 17.4% margin on $836K sales, the highest of any category.
- **Furniture** is a structural drag: 2.5% margin on $742K sales, with **Tables** and **Bookcases** losing money overall.
- **Discounting above 20% is destroying profit** — orders discounted 40%+ lose money at a -77% margin.
- The **West** region is the strongest performer (14.9% margin); **Central** is the weakest (7.9% margin).
- **Texas, Ohio, Pennsylvania and Illinois** are net loss-making states despite meaningful sales volume.
- November and December are the strongest sales months every year — a clear seasonal peak to plan around.

1. Sales & Profit Trend

Monthly sales are highly seasonal, with revenue consistently peaking in **November and December** of every year and dipping in Q1 (January–February). The overall trajectory is upward: total sales climbed from $484K in 2011 to $734K in 2014. The single strongest sales month in the dataset was **November 2014 ($112.3K)**, while the weakest profit month was **January 2012 (–$3.3K)**, a loss driven by heavy early-year discounting.

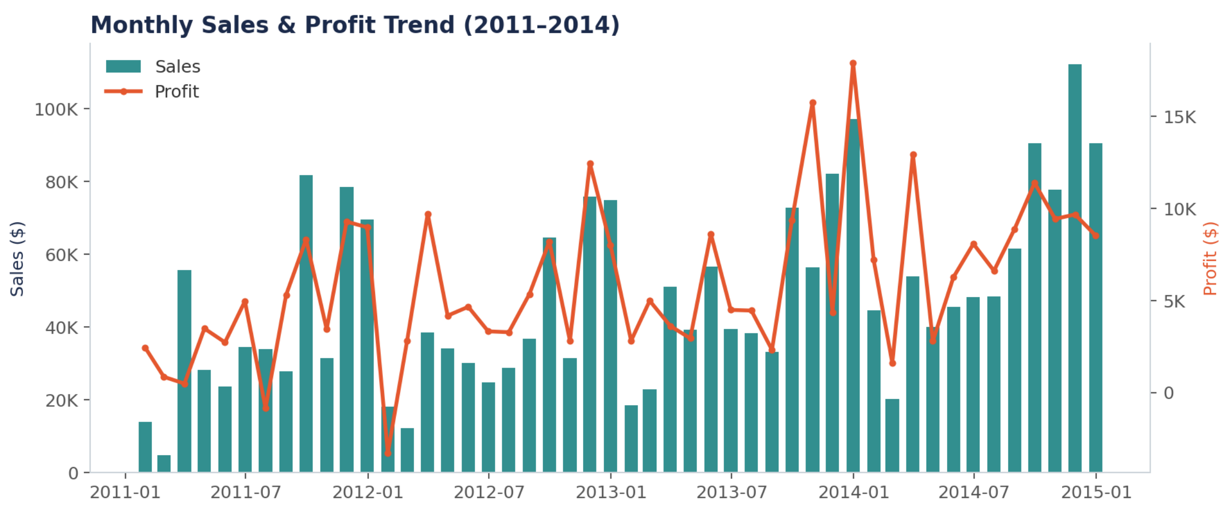

Figure 1. Monthly sales (bars) and profit (line), Jan 2011 – Dec 2014.

Year-over-Year Performance

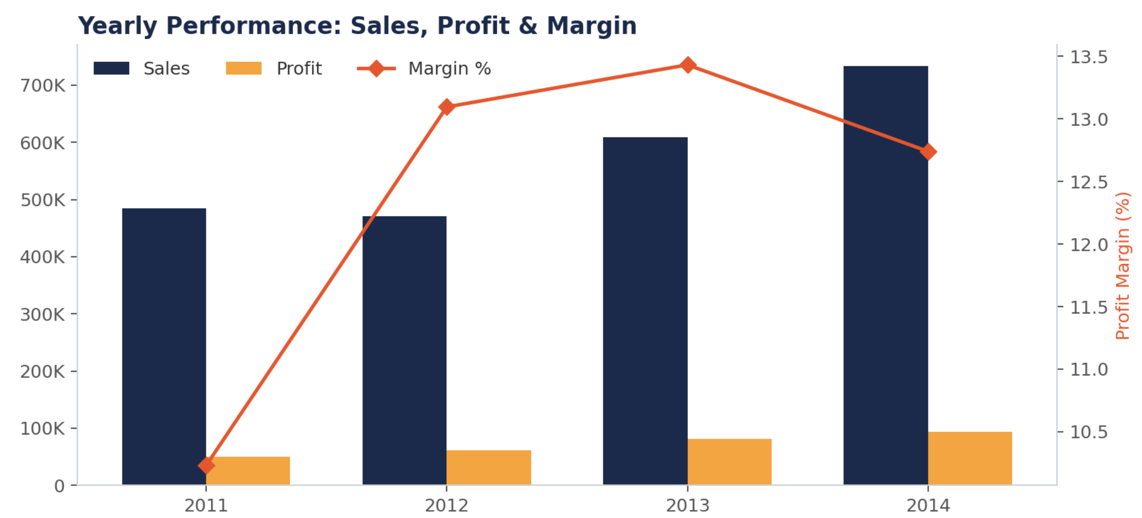

Figure 2. Yearly sales, profit and margin trend.

Year	Total Sales	Total Profit	Profit Margin	Sales Growth (YoY)
2011	$484,247	$49,544	10.2%	—
2012	$470,533	$61,619	13.1%	-2.8%
2013	$608,474	$81,727	13.4%	+29.3%

Year	Total Sales	Total Profit	Profit Margin	Sales Growth (YoY)
2014	$733,947	$93,508	12.7%	+20.6%

Insight: 2012 was a reset year — sales dipped slightly but margin improved to 13.1%, suggesting a deliberate pull-back on discounting. 2013 delivered the best margin of the period (13.4%) alongside the strongest growth (+29.3%). In 2014, sales grew fastest in absolute terms but margin slipped to 12.7%, hinting that growth was partly bought with discounts rather than pure demand.

2. Regional Performance

The **West** region leads on every metric that matters — sales, profit and margin — while **Central** lags well behind on profitability despite generating more sales than the South.

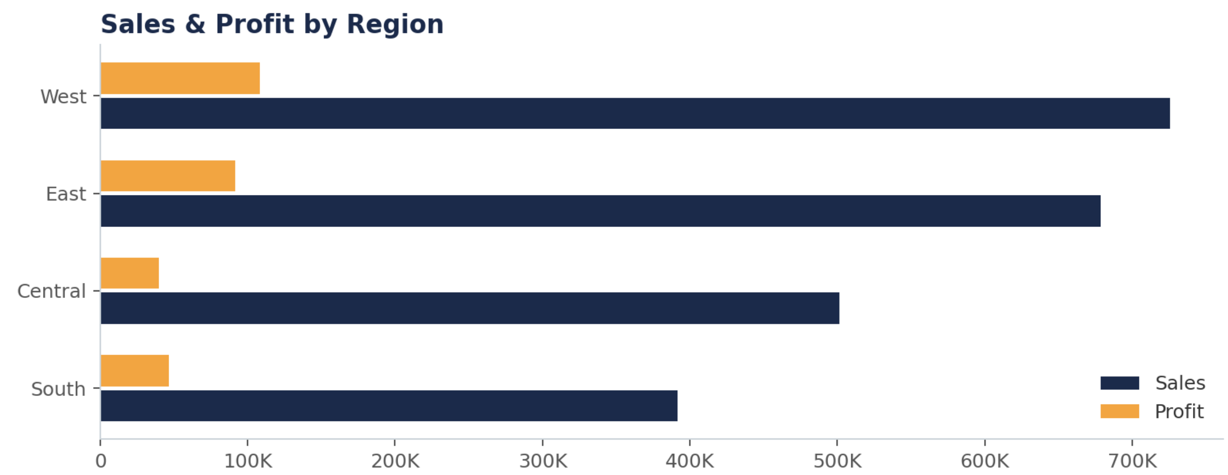

Figure 3. Sales and profit by region.

Region	Total Sales	Total Profit	Margin	Orders
West	$725,458	$108,418	14.9%	1,611
East	$678,781	$91,523	13.5%	1,401
South	$391,722	$46,749	11.9%	822
Central	$501,240	$39,706	7.9%	1,175

Loss-making states: at state level, the picture is more concerning. **Texas (–$25.7K), Ohio (–$17.0K), Pennsylvania (–$15.6K)** and **Illinois (–$12.6K)** are all net unprofitable despite six-figure sales volumes in Texas and Pennsylvania. This points to aggressive local discounting or an unfavorable product mix in these states, and warrants a dedicated pricing review.

3. Category & Sub-Category Analysis

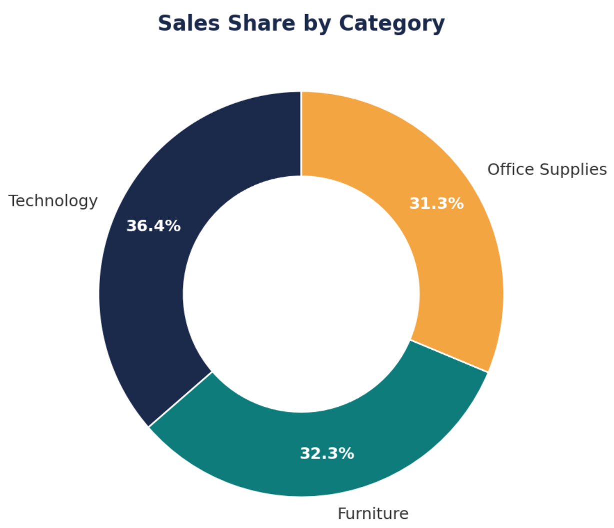

Figure 4. Share of total sales by category.

Category	Total Sales	Total Profit	Margin
Technology	$836,154	$145,455	17.4%
Office Supplies	$719,047	$122,491	17.0%
Furniture	$741,999	$18,451	2.5%

Sub-Category Deep Dive

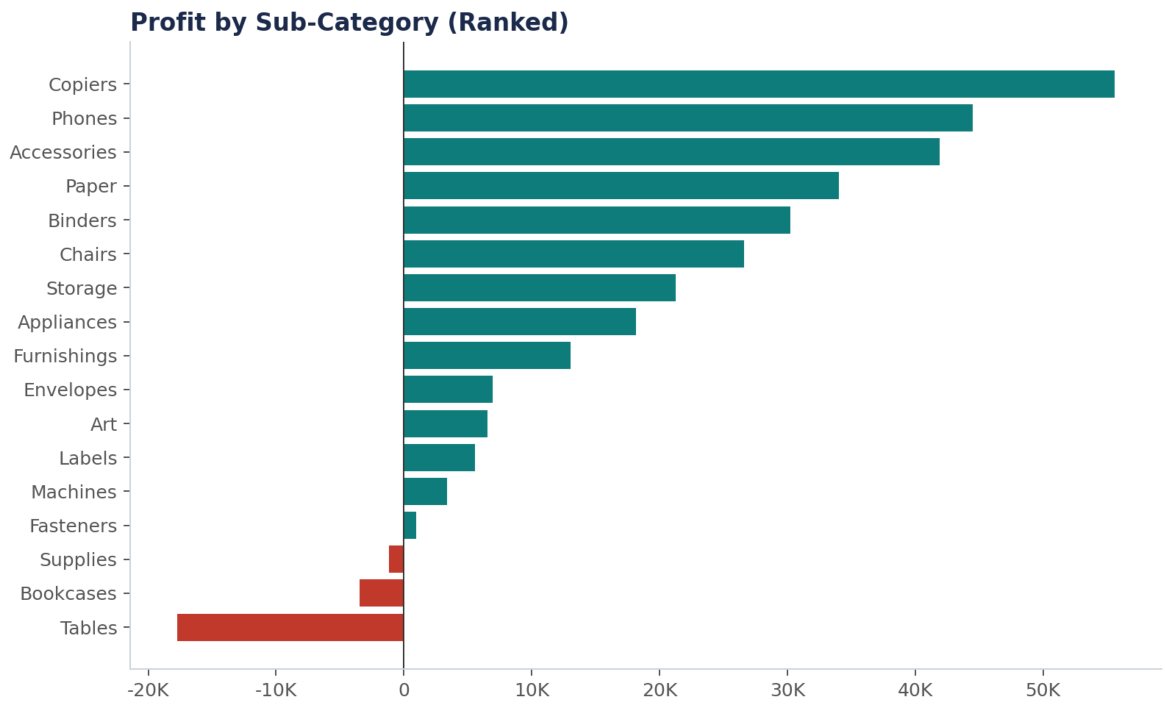

Figure 5. Total profit by sub-category, ranked lowest to highest.

Tables (–$17,725, -8.6% margin) and **Bookcases (–$3,473, -3.0% margin)** are the two sub-categories losing money overall — both sit inside Furniture, explaining that category's weak 2.5% margin. **Supplies** is marginally unprofitable too. On the other end, **Copiers (37.2%), Labels (44.4%) and Paper (43.4%)** are exceptionally profitable relative to their sales volume. **Furniture as a whole recorded a negative-profit month 17 times** across the four years — far more than Technology (4) or Office Supplies (2) — confirming this is a chronic, not seasonal, issue.

4. Discount & Pricing Impact

Discount level and profit are negatively correlated (r = –0.22) across the dataset, and the effect is dramatic once discounts move beyond 20%. Full-price orders (0% discount) run at a healthy **29.5% margin**; by the time discount reaches 21–40% the average order is **losing money (-15.3% margin)**, and discounts of 40%+ produce catastrophic **-77.4% margins**.

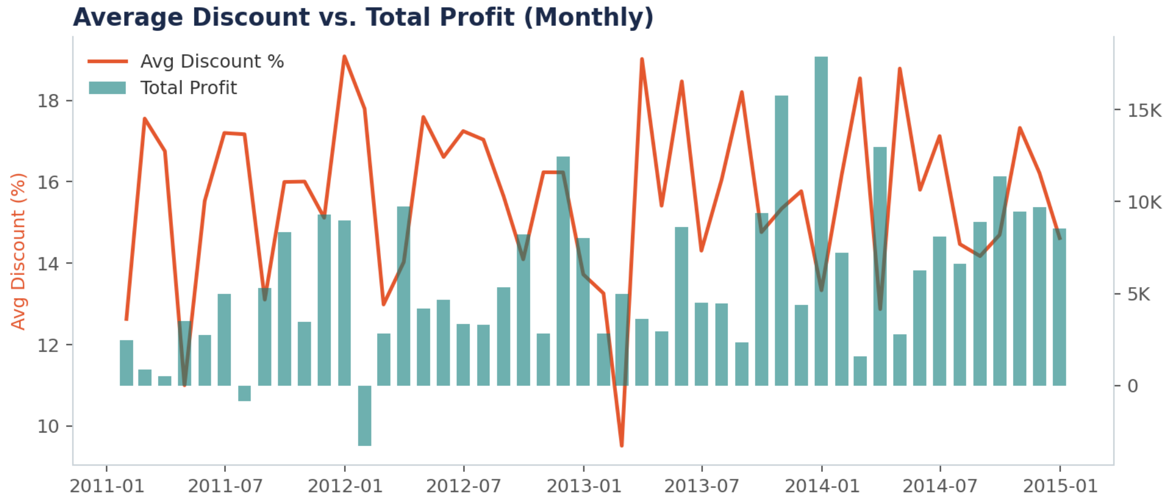

Figure 6. Average discount rate vs. total monthly profit.

Discount Band	Total Sales	Total Profit	Margin	Orders
0% (full price)	$1,087,908	$320,988	29.5%	2,644
1–20%	$846,522	$100,785	11.9%	2,507
21–40%	$234,138	-$35,817	-15.3%	400
40%+	$128,632	-$99,559	-77.4%	737

Insight: discounts above 20% should be treated as an exception, not a routine sales lever. The 737 orders discounted 40%+ collectively destroyed nearly $100K in profit — more than a third of the company's total profit for the entire period. Tightening discount approval thresholds is likely the single highest-leverage lever available to improve margin.

5. Customer Segment & Fulfillment

The **Consumer** segment drives the most volume ($1.16M sales, 409 customers) but converts it least efficiently (11.6% margin). **Home Office** is smaller but the most profitable per dollar of sales (14.0% margin), followed by **Corporate** (13.0%).

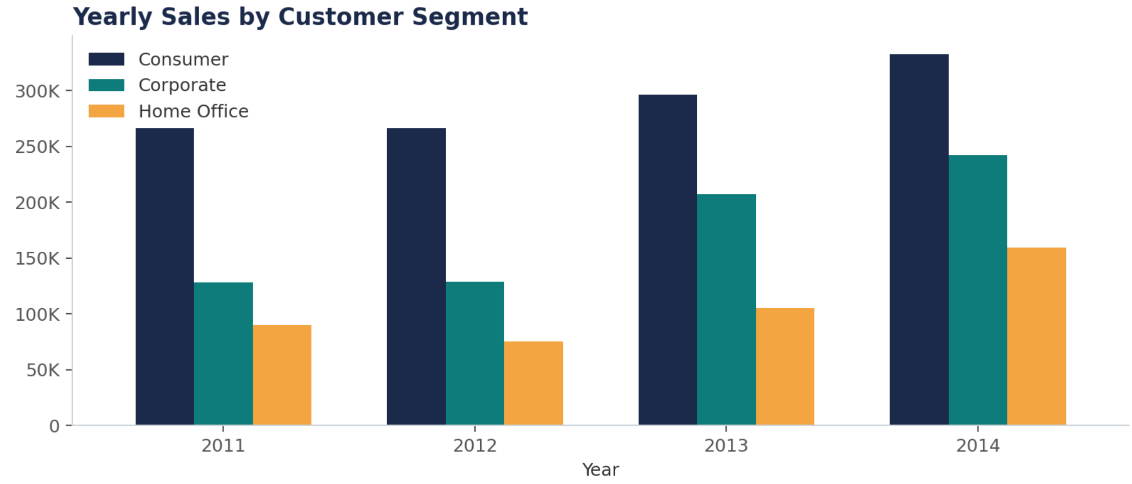

Figure 7. Yearly sales by customer segment.

Segment	Total Sales	Total Profit	Margin	Customers
Consumer	$1,161,401	$134,119	11.6%	409
Corporate	$706,146	$91,979	13.0%	236
Home Office	$429,653	$60,299	14.0%	148

By fulfillment method, **Standard Class** shipping accounts for the large majority of both sales ($1.36M) and profit ($164K), consistent with it being the default, lowest-cost shipping choice. Same Day shipping is a small niche (264 orders) but maintains solid margins, suggesting room to grow premium/expedited fulfillment as an upsell.

6. Product-Level Risk: Where the Losses Live

At the individual product level, **123 products generate above-average sales but still lose money overall** — these are the clearest candidates for repricing or discount-policy correction. The ten worst offenders by total profit are shown below.

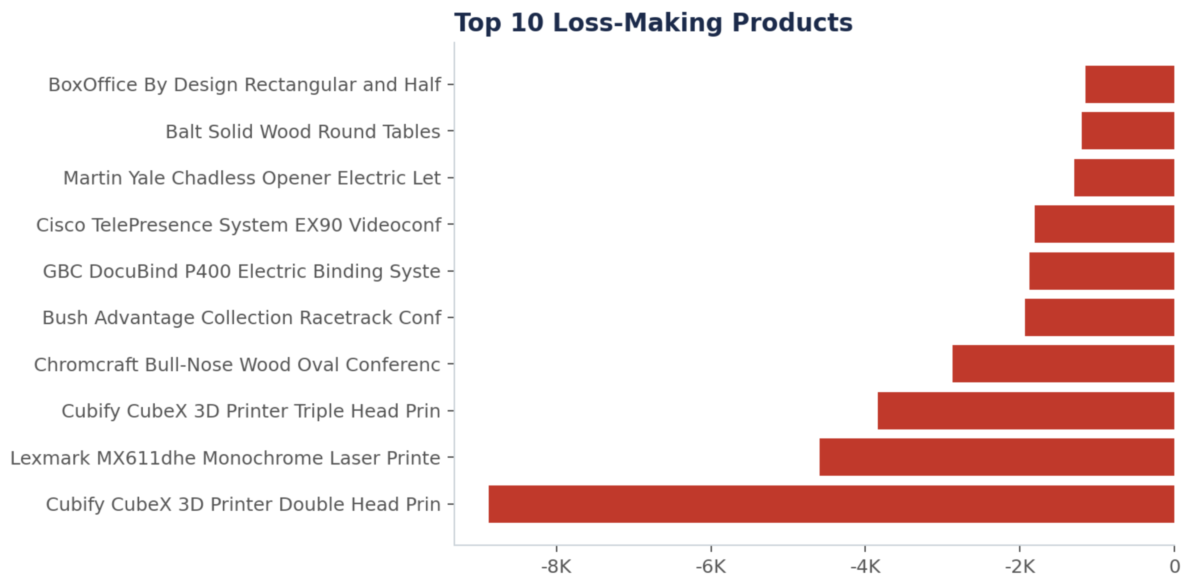

Figure 8. Ten products with the largest total losses.

Product	Total Sales	Total Loss
Cubify CubeX 3D Printer (Double Head)	$11,100	-$8,880
Lexmark MX611dhe Monochrome Laser Printer	$16,830	-$4,590
Cubify CubeX 3D Printer (Triple Head)	$8,000	-$3,840
Chromcraft Bull-Nose Wood Oval Conference Table	$9,918	-$2,876
Bush Advantage Collection Racetrack Table	$9,545	-$1,934
GBC DocuBind P400 Electric Binding System	$17,965	-$1,878
Cisco TelePresence System EX90	$22,638	-$1,811
Martin Yale Chadless Opener (Electric)	$16,656	-$1,299
Balt Solid Wood Round Tables	$6,519	-$1,201
BoxOffice Rectangular/Half-Moon Meeting Table	$1,706	-$1,148

Insight: the 3D printer line, the Cisco TelePresence system, and several conference/meeting tables stand out as high-ticket items sold at a loss — likely a combination of aggressive B2B discounting and thin baseline margins on premium equipment. Reviewing list prices and discount caps for these specific SKUs could recover a meaningful share of the $286K annual profit pool.

7. Recommendations

1. Cap discounts at 20%

Orders above a 20% discount consistently lose money. Introduce a manager-approval gate for any discount beyond this threshold, prioritizing Furniture and Central-region orders where the pattern is worst.

2. Fix Furniture profitability

Tables and Bookcases are structurally unprofitable. Re-price these sub-categories, renegotiate supplier costs, or reduce reliance on volume discounts to sell them.

3. Investigate loss-making states

Texas, Ohio, Pennsylvania and Illinois post negative profit despite strong sales. A regional pricing and freight-cost audit in these states is likely to surface quick wins.

4. Protect and grow Technology & Office Supplies

These two categories deliver ~17% margins. Prioritize marketing spend and inventory investment here over Furniture.

5. Plan inventory and staffing around Nov–Dec peaks

Sales peak reliably in November and December every year; ensure stock and fulfillment capacity are scaled ahead of this seasonal surge.

6. Review high-ticket loss SKUs individually

The 3D printers, TelePresence system, and premium conference tables are worth a manual pricing review rather than blanket discount policy alone.

Methodology & Sources

Analysis based on the Sample Superstore transactional dataset (9,994 order lines, Jan 2011–Dec 2014). Metrics were computed using SQL (aggregation, window functions for month-over-month growth and cumulative totals) and validated in Python (pandas). Profit margin is calculated as total profit divided by total sales for the relevant grouping. All currency figures are in USD and rounded to the nearest whole dollar unless otherwise noted.